Portuguese environmental policy at local level and Local Agenda 21 progress

The environmental politics at local level has been a major source of debate and thinking. To this much contributes the importance of local authorities[1] to deliver environmental services. Being in the majority of the world the governmental organization closest to people, they are in a privileged position to fulfil this function, as states the subsidiary principle and the chapter 28 of Agenda 21:

> "Because so many of the problems and solutions being addressed by Agenda 21 have their roots in local activities, the participation and cooperation of local authorities will be a determining factor in fulfilling its objectives. Local authorities construct, operate and maintain economic, social and environmental infrastructure, oversee planning processes, establish local environmental policies and regulations, and assist in implementing national and subnational environmental policies. As the level of governance closest to the people, they play a vital role in educating, mobilizing and responding to the public to promote sustainable development."

The same chapter provides the ground for the so called Local Agenda 21 (LA21):

> "Through consultation and consensus-building, local authorities would learn from citizens and from local, civic, community, business and industrial organizations and acquire the information needed for formulating the best strategies. The process of consultation would increase household awareness of sustainable development issues. Local authority programmes, policies, laws and regulations to achieve Agenda 21 objectives would be assessed and modified, based on local programmes adopted. Strategies could also be used in supporting proposals for local, national, regional and international funding."

Literature about the role of municipalities in attaining a sustainable development and about LA21 abounds, as readers of Environment & Urbanization are well aware of. Further information on this subject can be found elsewhere[2]. Moreover, there are already a number of articles describing LA21 success stories which provide a wealth of practical information[3].

In Portugal LA21 is still in its very beginning, though the Second European Conference on Sustainable Cities & Towns – which crated a great impetus towards the development of the European Sustainable Cities and Towns Campaign, with the approval of the Lisbon Action Plan – have taken place in the Portuguese capital in 1996. The same can be said about public participation in local or national policies. We report progress, probably as a consequence of international agenda, but there surely is a long walk to go through.

This article pretends, generally, to give an overall picture of the degree to which environmental concerns have been incorporated in local politics. We divided them into two fundamental stages of development: environment is embodied in local policies without any specific framework plan; and environment is embodied in local policies but is subject to a structuring plan such as a LA21 (which embraces social and economic concerns as well) or simply an Environmental Municipal Plan

(EMP). Most of the local authorities lie on the first branch. More precisely, the goals of this article are to:

- evaluate local environmental policies;
- report on the progress of LA21 in Portugal;
- broadly characterize environmental processes such as LA21 and EMP;
- determine the importance of these processes to change in local environmental conditions;
- identify barriers and possible facilitators to the development of those processes.

Environment in Portugal

Milestones in central government

Environmental matters in Portugal were first given some governmental support in 1971 with the creation of National Commission on Environment because of the forthcoming United Nations Conference on the Human Environment, Stockholm, 1972[4]. With it started the slow process of human qualification and information gathering. In 1974, the year of the democratic revolution, was created the Secretary of State of Environment – initially with little competencies but with time gaining political and practical relevance – headed by Arq. Gonçalo Ribeiro Teles, a famous landscape architect pioneer to bringing environmental concerns into public opinion who is, surprisingly, pretty much active. A major change occurred in 1985 with the creation of the Ministry of Planning and Natural Resources, from which the rebaptized Secretary of State of Environment and Natural Resources became dependent upon. This reinforced structure was given competencies in such areas as river management, pollution control and protected areas, together with a raised budget. The Environment Framework Law was published in 1987, forging the way for a whole set of laws and regulations devised to further develop environmental policies. Interestingly, the law is still in force.

Another landmark was the creation of the Ministry of Environment and Natural Resources, though the competencies on space planning were left in the Ministry of Planning and Natural Resources. More recently, in 1999, this serious error was solved with the integration of the two subjects in the Ministry of Environment and Territorial Planning. The last elected Government took up in March 2002 and introduced a great focus on cities and urban environment.

The political importance of the Ministry and its institutes has been increasing through the time, but it does not guarantee that the environmental values are fairly taken into account. Historical and structural dysfunctions as the ones concerning urban sprawl, space planning, water and air pollution, waste management and impact mitigation of highways and big dams are still poorly developed and have a lot to improve. Usually the problem is not so much the lack of regulation – which, in part as a result of the European Union directives, is well developed – but more of law enforcement, capacity building and political will.

Public participation is mostly limited to the environmental impact assessment processes and space planning plans. There are some signs of improvement in the discussion of policies and strategies, as the National Plan on Climate Change and the National Strategy on Sustainable Development

are examples. Unfortunately, what often happens is that public participation is only possible when plans are in their final phase of development, which obviously limits the possibilities of significant changes in those documents.

Competencies and basic description of local authorities

The 308 local authorities have a great deal of responsibilities in environmental policies, according to the competencies established by law[5], which include management and investment in the following domains:

Domain	**Relevant areas**
Water	• water supply • sewage collection and treatment
Waste	• municipal waste management
Air	• air quality supervision and maintenance
Nature and forest	• management of local protected areas • beach cleaning • river management inside urban perimeter • forest management
Space planning and urbanism	• Municipal Master Plan (MMP) production and approval • real estate projects appreciation and approval • proposal of site inclusion and exclusion to the National Ecological Network and to the National Agricultural Network
Housing	• social housing • urban renewal
Urban and rural equipment	• gardens • streets • markets
Energy	• low voltage distribution • public illumination
Transports and noise	• municipal roads • public urban transports • noise monitoring and reduction
Cultural heritage	•
Health	• health centres • support to disabled • old people
Civil protection	• municipal fire-fighters • combat and prevention of forest fires
Consumer protection	•
Industry and development	• tourism promotion • rural development • handcraft support • industrial and inert extraction licensing

Table 1 – Main environmental competencies of local authorities in Portugal.

Local authorities competencies usually involve inspection and appreciation of projects concerning the referred areas. Tough the responsibilities are many, most of them are in some way linked with central government (in Lisbon or through decentralized delegations), which can play a role in financial support, know-how and by issuing mandatory or advisory statements. In water and waste management the country if divided in associations of municipalities with common solutions, with some exceptions.

From the physical point of view there is a big disparity between municipalities (Table 2), the smallest being São João da Madeira, 25 km away from Porto, with a mere 8,3 km^2, and the largest being Odemira, in Alentejo, with 1720 km^2. There is still a lower level of administration called *freguesia* (similar to French *comunas*), but their responsibilities are extremely limited. As a result, what could be a privileged place to increase public participation and information exchange, is in fact little more than a bureaucratic administrative agency. The transference of powers from municipalities to *freguesias* is foreseen by the law but local authorities are renitent to do that since they would lose part of their direct control over people and territory. The discrepancy in *freguesias* distribution is also considerable: São João da Madeira has only one whereas Barcelos, in Minho, has 89, despite its area being only 378 km^2.

	Area (km^2)	**No. of *freguesias***	**Population density**
Largest	Odemira: 1720	Barcelos: 89	Amadora: 7597
Smallest	São João da Madeira: 8,3	São João da Madeira: 1	Alcoutim: 6,6
Average	298	14	294
Median	209	10	77
First quartile	122	5	30
Third quartile	383	18	177

Table 2 – Basic characteristics of Portuguese municipalities.

Many local authorities lack the necessary human means and budget to cope with such a demanding variety of duties. This often results in misconceived investments, technical errors, deficiencies and other faults that require further financial input. At the same time, it is not difficult to find investments one would consider, at first glance, highly dubious: it is the case of the spreading of luminous fountains and of traffic circles. It is clear that administration – at local and central levels – require a significantly more organized and planned management.

A selection of environmental indicators

To allow the reader having a global picture of environmental progress in Portugal, we have compiled the 1990 and 2000 values for a number of indicators (Table 3).

Indicator	**Value in 1990 (except otherwise stated)**	**Value in 2000 (except otherwise stated)**
Population	9 867 147 (1991)	10 356 117 (2001)

Gross national product		
Resident population with water supply (%)	77	89,7
Resident population with water drainage (%)	55	70,0
Resident population with water treatment (%)	21	55
Resident population with municipal waste collection (%)	95,2	98,4
Municipal waste production (kg *per capita*)	331 (1992)	453
Municipal waste recycled (%)	4,4 (1995)	3,4
Waste incinerated (%)	0,15 (1995)	11,7
Renewable electric energy production (excluding large hydroelectric: > 10 MW) (% of total electricity production)	4,6 (1995)	5,7
Passengers transported by train (%)	21,3 (1992)	19,6
Motorization rate (no. cars/1000 inhabitants)	205 (1992)	337
Territory covered by protected areas (%)	5,5	21,7
Emission of greenhouse gases (10^6 ton of CO_s equivalent)	60,6	77,0
Emission of tropospheric ozone precursors (kton of NMVOC equivalent)	811	1130

Table 3 – Portuguese basic environmental indicators and their evolution since 1990[6]

The trend of most indicators is linked to the significant economic growth Portugal felt during the 90s which, as would be expected, had significant environmental consequences, namely on urban sprawl (a result of internal migration and of immigration as well). The adoption of European Directives and international treaties binds the country to the accomplishment of several targets in such areas as water treatment, waste recycling, landfilling, renewable energy production, emission of greenhouse gases and of other air pollutants and air quality. There were also some improvements: the expansion of water system (supply, drainage and treatment), the closure of 328 waste dumps and the enlargement of territorial protected area (mostly because of Natura 2000 Network), among others.

Local environmental policy

To provide a broad picture of local environmental policy in Portugal a questionnaire was prepared and sent in October 2002 to all 308 municipalities with guarantee of confidentiality. Respondents were asked to choose between three options for each question or to write an open answer if none of the options applied (except yes/no questions about LA21). To qualify for statistical treatment, the closest option to the open answer was chosen. When this was not possible it was assumed there was no answer. Special commentaries will be made to provide detailed information and important remarks derived from open answers. A total of 155 questionnaires were received (slightly more than 50%).

The following graphs were based on this information; thus, “no answer” refers to the municipalities which have sent the questionnaire but have not answered a particular question.

Municipal waste management

Local authorities play a significant role in waste management. The country is currently divided into 35 waste management systems, each of them gathering several municipalities to stimulate common solutions. 75% of the questionnaires mention that selective waste collection is carried out in part of the territory, but the low percentage of recycling (about 3,4% of all municipal waste) indubitably shows the performance has to increase if the 2005 targets are to be met. Some municipalities already collect waste selectively all over their territory. Door-by-door selective collection is very low (13% of the answers) and surely restricted to urban centralities. The fragmented and sprawled nature of most of Portuguese urbanization in the last two decades is a serious drawback to implement it.

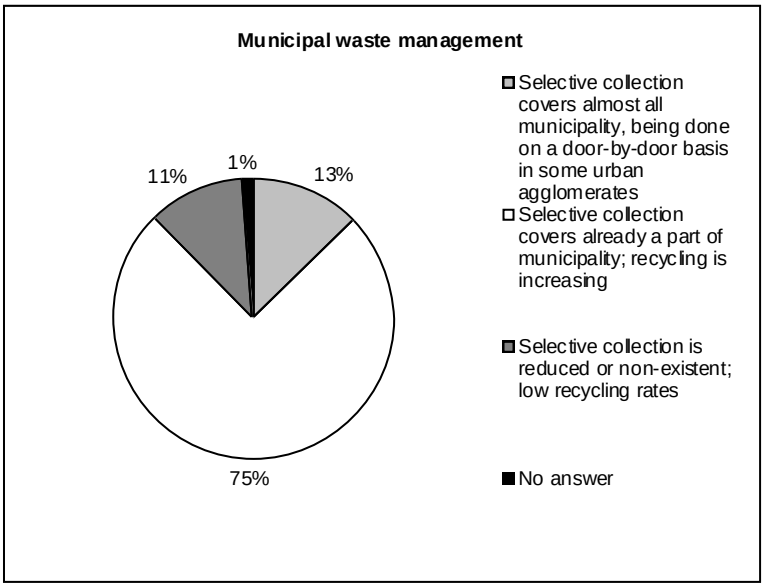

Figure 1 – Status of municipal waste policy.

Transportation

As a result of the significant expansion of highways, easy access to credit and the higher salaries Portugal has experienced since its adhesion to the European Union, among other reasons, there was a tremendous rise in the number of private cars. At the same time, public transports have received less investment. Nowadays panorama, with only 8% of local authorities stating their active promotion, is elucidative of this situation. It must be considered the particular situation of many rural municipalities which, because of their low population density and funds, lack public transports.

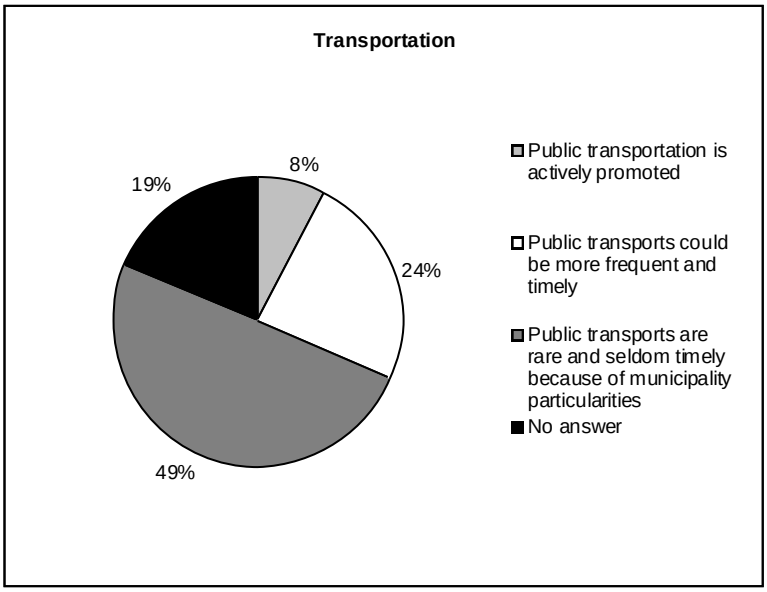

Figure 2 – Transportation policy.

Urban planning

In Portugal all municipalities are required to prepare MMPs. These plans set out the permitted land use and create a framework for urban planning in the territory; therefore, they confer local authorities a significant power to control land development. However, the little awareness they generally show to planning problems and their significant dependence over urbanization and housing taxes of new buildings to budget revenues, did not prevent several cities from "oil spill" growing[7]. The phenomena is more profound in coastal areas (Porto and Lisbon Metropolitanean Regions, Algarve) but affects interior cities as well (Vila Real, Guarda). MMPs were often designed to accommodate such growing, zoning as urban much bigger areas than would be required to cope with house demand. In spite of this, 35% of questionnaires mention that the territorial plan is carefully prepared according to environmental and urban concerns.

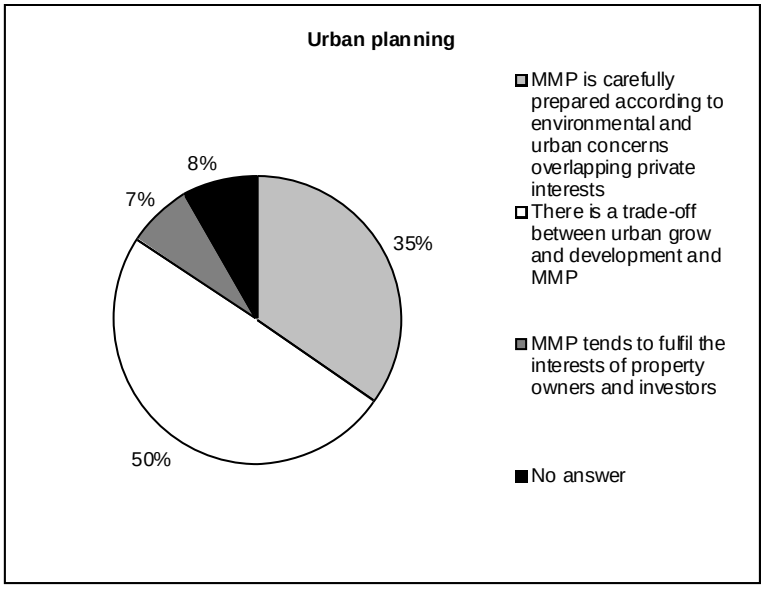

Figure 3 – Urban planning strategy.

Environmental education

Environmental education campaigns are markedly crucial if one wants to raise in the mid-long term society awareness. Intuitively this is an area which is growing fast. Only 5% of municipalities have stated little or no support for environmental education activities. Their next challenge is to prepare a coherent plan on a yearly basis so that educational activities have continuity and are based on a structured philosophy. Apparently, this is the case only for a minority of local authorities.

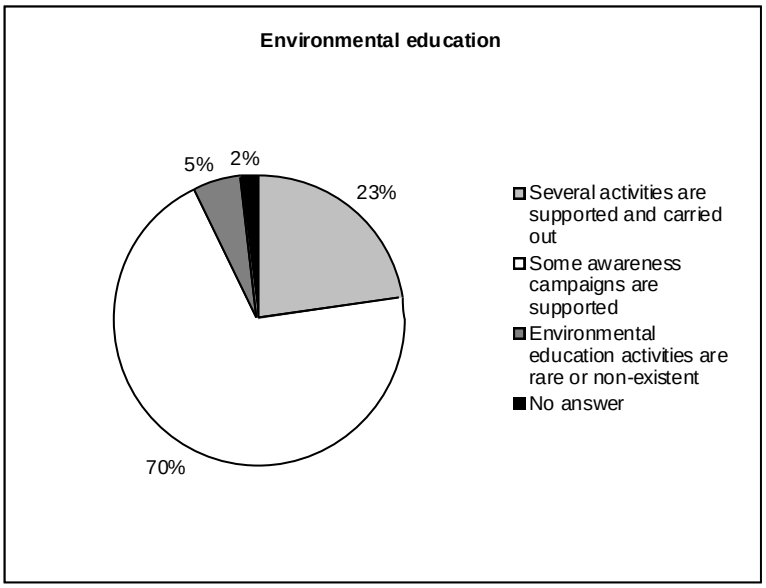

Figure 4 – Environmental education activities.

Public participation

Participatory democracy is not much developed in Portugal. Elected locals often interpret their role in a rigid way, as if public opinions were unnecessary since it is already represented through elections. Probably this is the consequence of a centralized country where, historically, little space was given to debate. There are reasons to believe this attitude is changing, since 27% of municipalities referred they actively promoted public participation and only 4% declared the opposite.

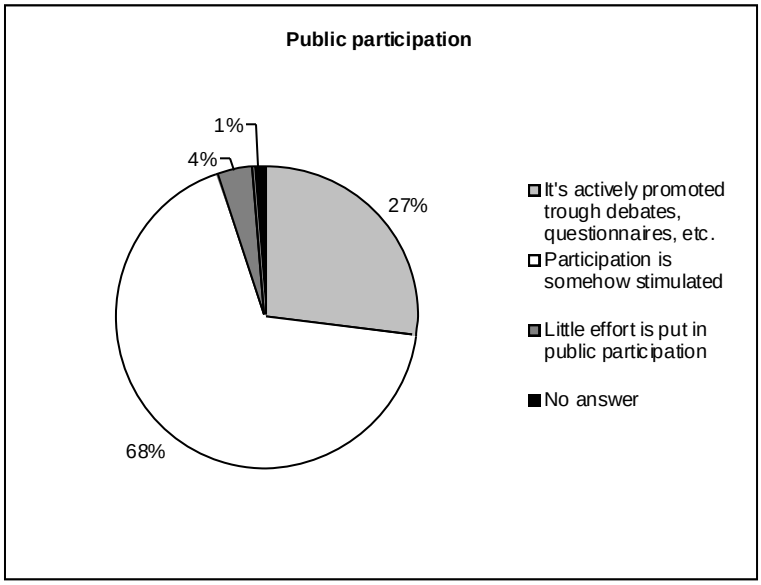

Figure 5 – Effort put in public participation.

Access to information

When in contact to local authorities one usually feels some difficulty in accessing some administrative documents, though it is granted by law. The reasons pointed before also explain this situation, but again the questionnaire does not confirm this trend. Instead, it suggests that access to information is easy (78% of respondents). Sometimes the problem is not so much asking for a precise document, but more knowing which information is available. This requires the presence and support of a technician, who can have little spare time. Apparently the attitude towards freedom of information is getting better, but to foster an effective policy in this matter deeper measures are necessary.

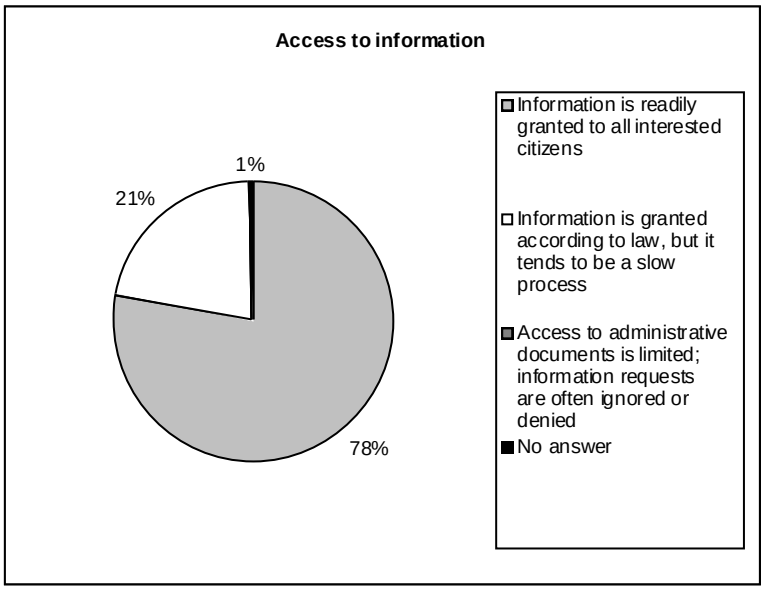

Figure 6 – Freedom of access to information.

Local Agenda 21

The general questionnaire included questions about LA21. The survey shows that the knowledge about the concept is high (83%) – though it is not possible to assess to which degree. This figure is significantly higher than the 48% found by Carter and Nunes da Silva (2000)[8] in 1997. It also points out that a huge majority of municipalities thinks it would be useful to develop such a process (89%) and, surprisingly, that 68% intents to start it during the present mandate.

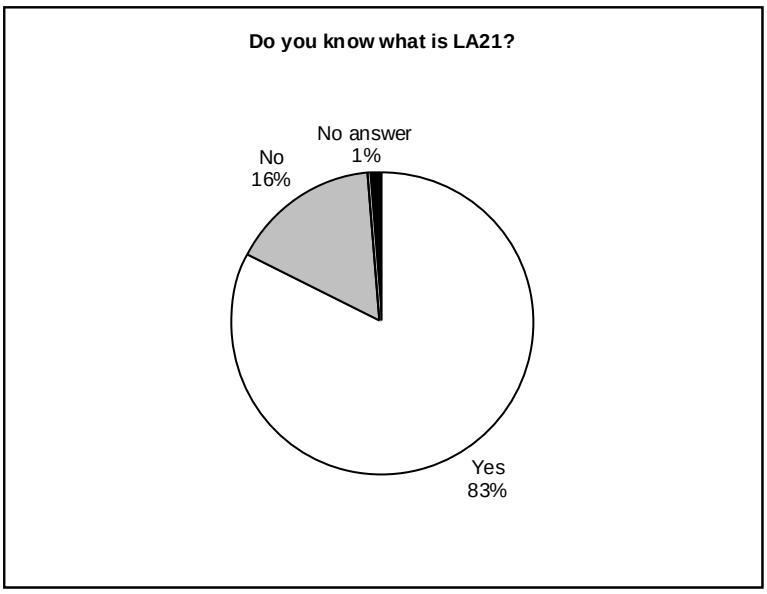

Figure 7 – Knowledge about LA21

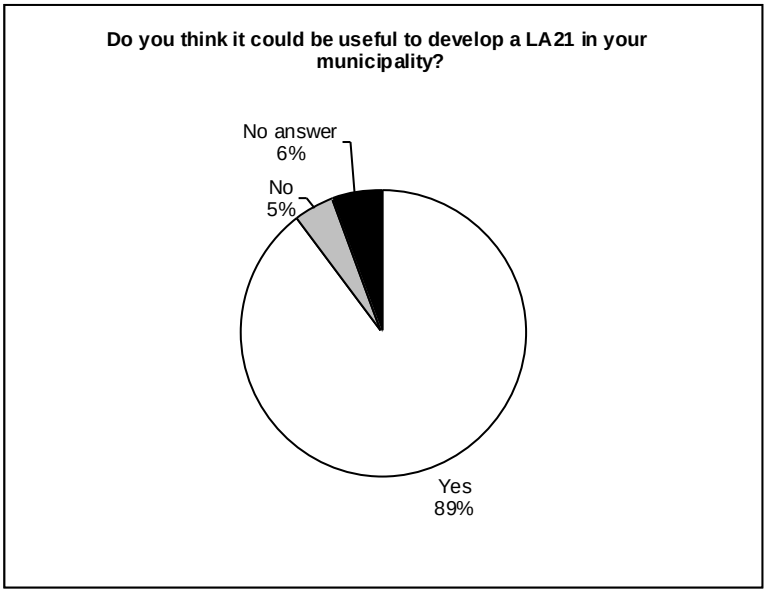

Figure 8 – Usefulness of LA21 for local authorities

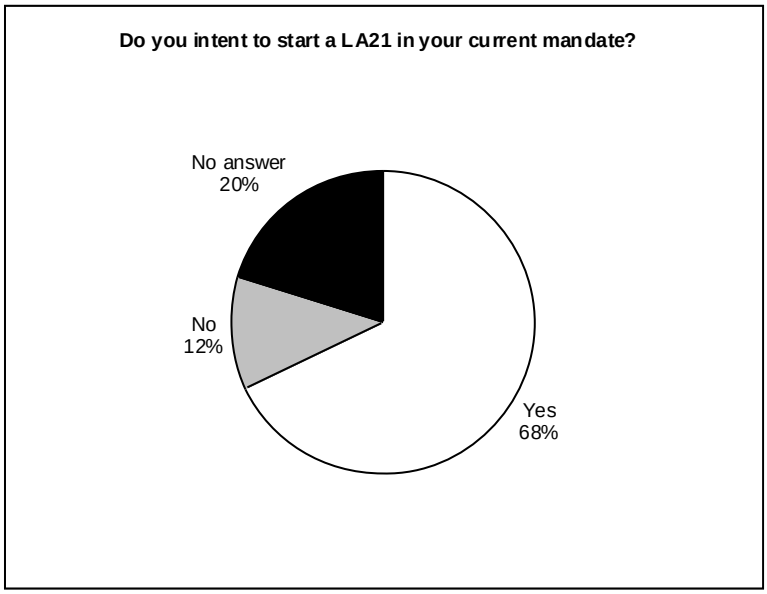

Figure 9 – Intentions about starting LA21.

Local environmental processes

In addition to the general questionnaire a more detailed one about local environmental processes was also sent. Because LA21 is still in its very beginning in Portugal, as the survey will show, we preferred to broaden the scope of it and include EMPs and other processes. EMPs are more focused in environmental concerns, whereas LA21 cover social and economic issues as well. The level of participation attained in a LA21 is also substantially higher than in an EMP.

A total of 18 questionnaires were received, representing 11,6% of all answers received from the general survey. That is, around 12% of the respondents have some sort of environmental process. The figure for the whole country will probably be lower, since municipalities with environmental processes are more

prone to answer than the others. Carter and Nunes da Silva (2000)[9] refer 17%, but it surely is an overestimation.

LA21 is in its initial stages, as can be inferred from ?. In fact, only one municipality (6%) has started – quite recently – such a process (though others local authorities claim that, but since those processes do not comply with United Nations guidelines they were considered EMPs). Time will tell whether this is a true LA21 or not. EMPs represent the majority of environmental processes, probably because of their apparently less quarrelsome nature. We considered EMP+ the processes that are somewhere between a LA21 and a EMP. The Lisbon and Tagus valley and North regions (NUTS II) dominate in number of ongoing processes, with 7 and 5, respectively. Follows Alentejo (3), Algarve (2) and Center (1). The poorest regions, Azores and Madeira archipelagos, have none. Most of the processes are very recent: 44% started in 2002, 39% between 1999 and 2001 and only 18% dates back to 1998 or before.

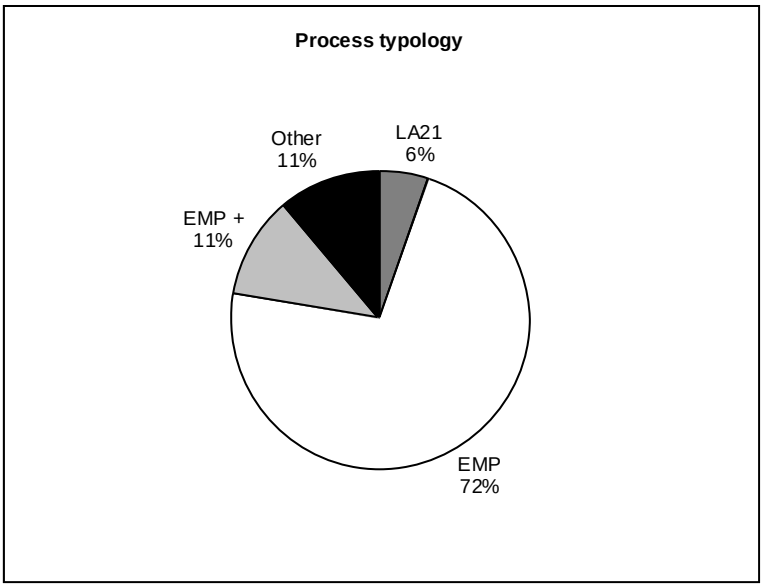

Figure 10 – Typology of ongoing processes (percentage relatively to the total number of processes).

Articulation with other municipal policies

Only 33% of the processes are restricted to the environmental department, the other 67% covering all municipal policies. This is very important to prevent it from being a mere “whim” of some technicians to become strategy born in mind by all departments. Several measures were taken to ensure this articulation (?). However, the very high rate of nonrespondents suggests that most of local authorities did not adopt any special measures regarding interdisciplinarity.

State of process implementation

LA21 was divided into the seven main phases recommended by ICLEI[10]:

- Phases 1 and 2: Activation and organization
- Phase 3: Participation Forum
- Phase 4: Problem analysis
- Phase 5: Action Plan definition
- Phase 6: Action Plan implementation
- Phase 7: Monitoring

Figure 11 – Measures taken to ensure articulation with other municipal policies (percentage of the total number of

To facilitate data interpretation, the state of implementation of all processes, including EMPs, was characterized according these steps. Please be aware that the phases are supposed to be sequential, but because most processes differ substantially from LA21 – many being *sui generis* enough to be labeled "other" – this is often not the case. Phases were further grouped into three categories likewise has been done in the Italian LA21 survey[11] (Table 4).

Beginners	Intermediate	Pioneers
Phase 1: Activation Phase 2: Organization	Phase 3: Participation Forum Phase 4: Problem analysis	Phase 5: Action Plan definition Phase 6: Action Plan implementation Phase 7: Monitoring

Table 4 – Main classes of process implementation.

Figure 12 – State of process implementation (percentage of the total number of processes).

Figure 13 – State of process implementation by kind of process (absolute number of processes are shown).

The questionnaire did not distinguish between phases 1 and 2. As a result, we considered all respondents to have achieved both of them. Participation Forum, as would be expected from the low number of ongoing LA21, has been put in practice in a mere 11% of the cases. Problem analysis is a consensual phase: 78% have carried it out while the remaining 22% plan to. 50% of the processes have already completed Action Plan definition but only 39% of the rest intent to prepare one. Surprisingly, some processes will not go beyond problem analysis, which is obviously inadequate if they are to be effective. Action Plant implementation is reported by 28%, confirming the idea that most of the processes remain on a theoretical basis and rarely know any practical development. Monitoring seems to have been completely left out or, perhaps, time has not come yet to start it.

It is absolutely astonishing that there is not only one ongoing LA21 in an intermediate level. World leaders called to LA21 implementation in 1992; yet, 11 years have passed and the progress can barely be seen.

Lisbon and Tagus valley is the richest region and is also the one where environmental processes are most developed (?).

Figure 14 – State of process implementation by region (percentage of the total number of processes).

Organisational structures

One of the main organisational tasks is the set up of a co-ordinating group responsible for following the whole process and adopting the required measure to ensure its applicability and efficiency. It is usually composed of high level municipal staff and strategic partners (it is mainly a political body). The co-ordinating group was set up in 66% of the processes, 17% intent to create it and the other 17% have not foreseen the group.

Another important step is the development of partnerships with various institutions. This puts in practice the subsidiary principle and promotes the sharing of responsibilities. It is commonly regarded as an excellent way to stimulate public participation and its involvement in local sustainable development. Unfortunately, perhaps because some local authorities fear to lose some power with this kind of partnerships, they are not common and were established solely in 28% of the cases. 55% of the respondents declared partnerships are still to be agreed and the remaining 17% will not implement them.

Public participation

Environmental processes are deeply rooted in public participation, since environmental protection and awareness raising are interconnected. Getting people to actively participate is no easy task and this difficulty is reinforced by a state of general apathy the population has been driven for the last decades. When asked about the methods used to stimulate participation more than half of the questionnaires were left blank, which may be an indication of the little effort it is still put in reverting the situation described. Press was reported by 44% of the respondents followed by workshops and questionnaires (28%).

Figure 15 – Methods used to stimulate public participation (percentage of the total number of processes).

Action Plan structure

Another focus of the questionnaire was the structure adopted for the Action Plant. Because only 9 processes have completed this phase, the results are based on this limited set of data. All plans include objectives and targets, but when we come to implementation mechanisms the results are not so encouraging. In fact, only 33% (3 Action Plans) have completed an implementation calendar and progress indicators are reported by no more than 44%. Moreover, the actions of almost half of the plans lack a reliable cost estimation. This scenario suggests that existing plans are not of much use, though they can still be used as a guiding instrument and policy reference tool.

Action Plan scope

By definition, a LA21 must embrace environmental, social and economical areas. Although EMPs are more environmental focused and dominant in Portugal, most of them (44%) followed that philosophy and only 11% were in fact restricted to the field. Policy integration is a fundamental step in local sustainable development.

Figure 17 – Action Plan structure (percentage of the total number of Action Plans).

Figure 16 – Action Plan scope (percentage of the total number of Action Plans).

Action Plan implementation

If the number of Action Plans is low, even lower are the statistics of their implementation. Some processes are recent enough for this to be acceptable, but others have started long ago and should be more advanced. From the direct contact with some municipalities it was clear that the political will necessary to keep the work on has stalled, bringing the process at least to an interruption (this led some local authorities to review the plan even before putting into practice the

one they have). In addition, 11% of the municipalities have explicitly stated their Action Plan will not be carried out.

Globally, LA21 implementation (that is, the transition of Agenda 21 principles to practice) seems far from being a reality.

Figure 18 – Action Plan implementation (percentage of the total number of Action Plans).

[1] In Portugal the lowest level of administration with significant powers are municipalities, so both terms will be considered synonyms in this article.

[2] Reference materials about LA21 include ICLEI (1996), *The Local Agenda 21 planning guide: an introduction to sustainable development planning*, ICLEI, Toronto; Cotter, B. and Hannan, K. (Environs Australia) (1999), *Our Community our Future: a Guide to Local Agenda 21*, Commonwealth of Australia, Camberra; Lawrence, Gary (1998), *The future of Local Agenda 21 in the new millennium*, Millennium papers: issue 2, UNED-UK, Londres; ICLEI (2002), *Second Local Agenda 21 survey*, Background paper no. 15, United Nations Department of Economic and Social Affairs; ICLEI (1998), *Barriers to the implementation of Local Agenda 21*, ICLEI, Toronto.

[3] Suggested articles include Miranda, Liliana and Hordijk, Michaela (1998), *Let us build cities for life: the national campaign of Local Agenda 21 in Peru*, Environment and Urbanization, vol. 10, no. 2; Roberts, Ian (2000), *Leicester environment city: learning how to make Local Agenda 21, partnerships and participation deliver*, Environment and Urbanization, vol. 12, no. 2; Wild, Andrew and Marshall, Robert (1999), *Participatory practice in the context of Local Agenda 21: a case study evaluation of experience in three english local authorities*, Sustainable Development, vol. 7: 151-162; Jörby, Sofie (2002), *Agenda 21 in the municipalities: a tool towards sustainability*? Journal of Environmental Planning and Management, vol. 45 , no. 2: 219-244.

[4] For more information please refer to Melo, João Joanaz de and Pimenta, Carlos (1993), *O que é: ecologia*, Difusão Cultural, Lisboa.

[5] Law no. 159/99, 14th September 1999.

[6] Sources: Instituto Nacional de Estatística (http://www.ine.pt); Ministério das Cidades, Ordenamento do Território e Ambiente (2002), *Relatório do estado do ambiente 2001*, MCOTA, Lisboa; Direcção-Geral do Ambiente (2000), *Relatório do estado do ambiente 1999*, DGA, Lisboa; Direcção-Geral dos Transportes Terrestres (2002), *Evolução do sector de transportes terrestres – documento de síntese (1992-2000)*, DGTT, Lisboa; Direcção-Geral de Energia (http://www.dge.pt); Eurostat (http://europa.eu.int/comm/eurostat).

[7] A deep analysis of urban planning in Portugal can be found at Carter, Norma and Nunes da Silva, Fernando (2001), *Recent changes in territorial planning and the system for controlling urban development in Portugal*, Town Planning Review, vol. 72, no. 3: 341-370.

[8] Carter, Norma and Nunes da Silva, Fernando (2000), *Local Agenda 21: progress in Portugal*, European Urban and Regional Studies, vol. 7, no. 2: 181-186.

[9] Refer to previous note.

[10] Check reference 2.

[11] Focus Lab (2002), *Local Agenda 21 in Italy 2002: survey on Local Agenda 21 processes implementation in Italy*, Focus Lab, Modena.